INCH-POUND

MS500040K
17 November 2011
SUPERSEDING
MS500040J
1 March 1991

DETAIL SPECIFICATION SHEET

TRANSMITTER, LIQUID QUANTITY, FLOAT TYPE, ENGINE FUEL

Inactive for new design after 28 September 1993.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-10986, "Indicators and Transmitters: Liquid, Temperature and Pressure".

INTENDED USE:

1. An indicator (dashboard unit) and a transmitter (sending unit), when connected by a single conductor cable (not supplied) and properly grounded, comprise a liquid quantity gauge. This device is used on military motor vehicles and other internal combustion applications to inform the operator of the level of fuel in the supply tank or tanks.

REQUIREMENTS:

1. Material: Material shall be in accordance with unit 2, type I, class 1, of MIL-DTL-10986.

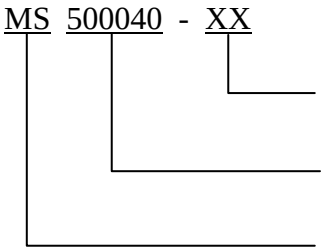
2. Finish: Finish shall be in accordance with unit 2, type I, class 1, of MIL-DTL-10986.

3. Design: Design shall be in accordance with unit 2, type I, class 1, of MIL-DTL-10986.

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NOTES:

1. The part or identifying number to be used for the fuel gauge transmitters acquired to this specification is created as follows:



Example: MS500040-10

Dash number (see table I)

Specification number

MS Prefix

TABLE I. Fuel quantity transmitters dash numbers.

Dash number	Former part number	Dimensions				Float position	Float type
		A Max (inch)	B Min (inch)	C (inch)	R Max (inch)		
-1	10872707	7.23	0.13	3.96 3.48	3.93	I	Regular
-2	8376496 7526038	7.67	0.39	3.96 3.48	4.15	I	Regular
-3	8376495	10.28	0.47	5.42 5.11	5.51	II	Regular
-4	10942801	11.94	0.47	5.50 5.20	8.50	II	Regular
-5	7034597	15.56	0.29	8.21 7.90	9.32	I	Regular
-6	8376491	16.90	0.19	8.21 7.90	10.53	I	Regular
-7	10921745	17.15	2.35	9.77 9.71	9.03	II	Regular
-8	7064625	33.50	1.40	14.79 14.49	21.07	I	Long
-9	8737838	14.59	0.53	7.40 7.00	8.67	I	Regular
-10	12332788	11.65	3.19	7.47 7.41	4.47	I	Regular

2. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

3. Unless otherwise specified, issues of referenced documents are those in effect at the time of solicitation.

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4. When tested in accordance with MIL-DTL-10986, resistance calibration measurements shall be 0.00 to 0.05 ohms at the "EMPTY" position, and 29.50 to 31.50 ohms at the "FULL" position, (see figure 1 and table I).

5. Any portion of the assembly that is below the mounting plane (see figure 1) shall pass through an opening of 1.562 inches in diameter.

6. When testing on the applicable fuel tank, a fuel reserve of 10 percent of the applicable fuel tank capacity shall remain when the float reaches the "EMPTY" position (see figure 1) with fuel tank level declining.

7. The float lever fulcrum located by dimension C shall be established in line with the 50 percent fuel level reference (see figure 1 and table I) when testing on the applicable fuel tank.

8. The MS part number and manufacturer's identification shall be marked on the body of the transmitter in accordance with MIL-STD-130, "Identification Marking of U.S. Military Property".

9. The transmitter shall have a steady, continuous output for all float positions and shall track linearly with the float position.

10. Unless otherwise specified, all steel parts except stainless steel shall be zinc plated in accordance with type II, service condition I, of ASTM B633, "Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel", in accordance with MIL-DTL-10986.

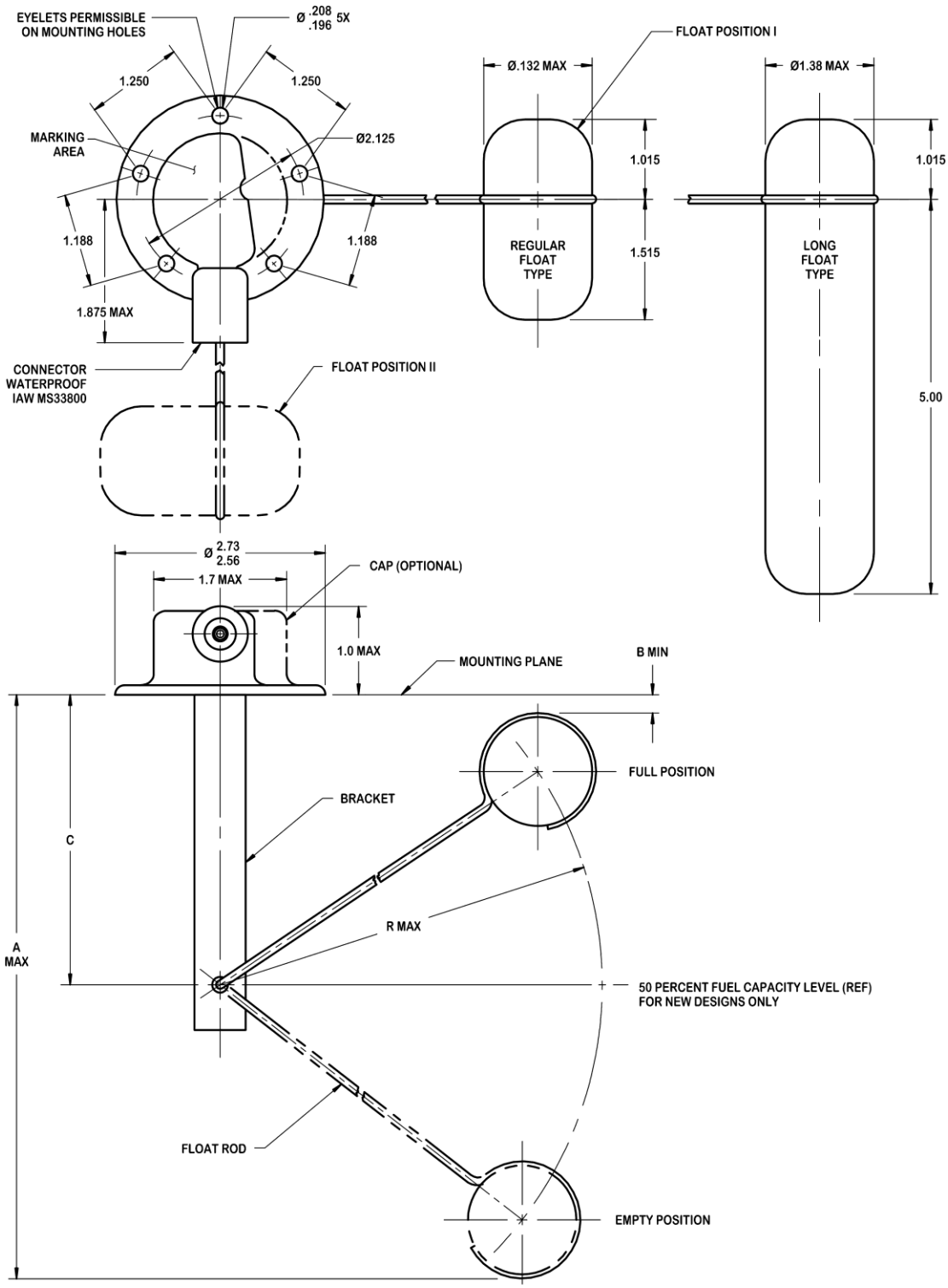
11. The material used for the float shall be brass, non-leaded sheet or strip copper alloy UNS number C26000 or C26800, annealed temper, in accordance with ASTM B36/B36M, "Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar".

12. The use of recycled materials that meet the requirements of the applicable material specifications and that do not jeopardize the intended use of the item is strongly encouraged.

13. For the associated indicator (fuel gauge), see MS24544, "Indicator, Liquid Quantity: 24 Volt DC, Engine Fuel".

14. This standard is not intended to limit construction to features other than the dimensions, notations, and references contained within.

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NOTES:
1. All dimensions are in inches.
2. Unless otherwise specified, tolerances ±0.015.

FIGURE 1. Fuel quantity transmitters.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
DLA - GS

Preparing Activity:
DLA - GS8

(Project 6680-2011-002)

Review Activity:
Army - CR4

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